

Tutorial 2

ECON 101 – Macroeconomics

Fall 2025

Instructions

Answer the following questions carefully. Show all calculations and provide brief explanations where required. Use correct units (billions, millions, etc.).

Problem 1: Calculating GDP with the Expenditure Approach

Suppose the Canadian economy in 2024 has the following data (in billions of dollars):

- Consumption (C) = \$1,800
- Investment (I) = \$600
- Government purchases (G) = \$720
- Exports (X) = \$450
- Imports (M) = \$520

- a) Calculate nominal GDP using the expenditure approach.
- b) Suppose the population is 40 million. Calculate GDP per capita.
- c) If potential GDP is \$3,800 billion, compute the output gap as a percentage of potential GDP.

Problem 2: GDP from the Income Side

You are given the following data for Canada (in billions of dollars):

- Wages: \$1,500
- Rent: \$200
- Interest: \$250
- Profits: \$400
- Indirect taxes (less subsidies): \$250

- Depreciation: \$300
- a) Compute Net Domestic Income at factor cost.
 - b) Add indirect taxes (less subsidies) and depreciation to compute GDP at market prices.
 - c) Compare your answer with Problem 1 and explain the relationship between expenditure and income measures of GDP.

Problem 3: Nominal vs. Real GDP and Price Indexes

The following data describe the economy's production of two goods (apples and laptops):

Year	Quantity of Apples	Price of Apples	Quantity of Laptops	Price of Laptops
2022	1,000	\$2	200	\$1,000
2023	1,200	\$3	220	\$1,100
2024	1,300	\$3	250	\$1,200

- a) Compute nominal GDP for each year.
- b) Using 2022 as the base year, compute real GDP for 2022–2024.
- c) Calculate the GDP deflator for each year and interpret what it measures.
- d) What is the inflation rate between 2023 and 2024, as measured by the GDP deflator?

Problem 4: Underground and Informal Economy

Canada's measured GDP in 2024 is \$2,900 billion. Economists estimate that the underground economy accounts for 7% of measured GDP.

- a) Compute the actual size of the underground economy in billions of dollars.
- b) Adjust measured GDP to include the underground economy.
- c) If the population is 39 million, what is the difference in GDP per capita between measured GDP and adjusted GDP?
- d) Briefly explain why excluding the underground economy can distort international GDP comparisons.

Problem 5: Chain-Weighted Real GDP

Consider the following data for a simplified economy that produces only coffee and textbooks:

Year	Quantity Coffee	Price Coffee	Quantity Textbooks	Price Textbooks
2023	500	\$4	100	\$60
2024	600	\$5	120	\$70

- Compute nominal GDP for 2023 and 2024.
- Compute real GDP in 2024 using 2023 prices.
- Compute real GDP in 2023 using 2024 prices.
- Calculate the chain-weighted real GDP growth rate between 2023 and 2024.
- Explain why the chain-weighted method is preferred over using a fixed base year.

Problem 6: GDP and Economic Well-Being

Suppose two countries (Country A and Country B) have the same GDP per capita of \$40,000. However, the data show:

- Country A works on average 2,200 hours per worker per year.
 - Country B works on average 1,600 hours per worker per year.
- Compute GDP per hour worked for each country.
 - Which country has higher productivity?
 - Discuss why GDP per capita alone might not fully capture differences in well-being.